BEACON LABS

CYBERSECURITY EDUCATION LABS

Workshop

BOISE STATE UNIVERSITY

COLLEGE OF ENGINEERING

Department of Computer Science

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Setup

Android Studio Installation

Steps:

1. The latest versions of Android Studio have since deprecated the API version used in the workshop. To download a compatible version of Android Studio, open the Android Studio Download Archive (full link at the bottom of this step) and find “Android Studio Ladybug Feature Drop | 2024.2.2 Patch 2” from February 26, 2025. Expand the drop down to download an installer if you’re on Windows or the zip archive if you’re on Linux. For the Windows installer, the standard options will suffice. For Linux, after downloading and untarring the archive, cd into the untarred directory and then into the bin directory and run the `studio.sh` script to start Android Studio.

   https://developer.android.com/studio/archive

2. Once the installation has finished, open Android Studio.
3. The first time Android Studio is opened, it may prompt you to agree to their terms and conditions and install their SDKs, step through it for a standard installation if prompted.
4. Once everything has finished downloading, move to the next section.

Emulation Device Setup

Steps:

1. The device we’ll be using for this activity is the Pixel 5 with API 28 (“Pie”). To start creating our emulation device, first download the activity starter code from this Google Drive link.

   https://drive.google.com/file/d/17KuEH9aQtD4UuVXMFIOAWq97-_Z-CzXk/view?usp=sharing

2. Once downloaded, unzip into a directory.

3. In Android Studio, choose the "Open" option (Figure 1) and open the directory where the activity materials were saved to. Depending on the tool used to unzip the project it may create a new directory with a subdirectory. Use the subdirectory as the directory to open (Figure 2).

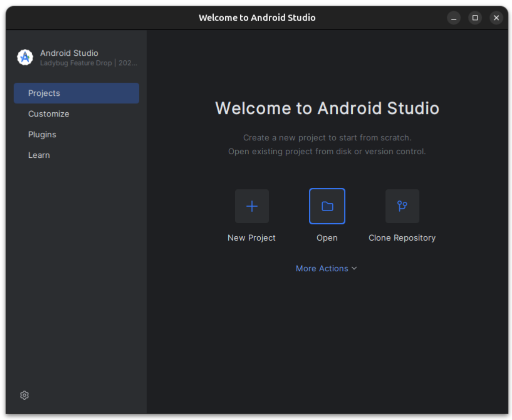

Figure 1: Android Studio open project

4. Once the project has been opened, navigate to the Device Manager (Figure 3), click the "+" button and select the "Create Virtual Device" option.
5. Scroll down to or search for the "Pixel 5" device, select it and click Next (Figure 4).
6. For the System Image, under "Recommended" scroll down until you see "Pie" with an API version of "28." If it isn't downloaded already, click the download icon next to "Pie" and agree to the SDK license agreement to download the system image. See Figure 5 to verify that the correct system image has been selected.
7. Once downloaded, select Next then Finish before moving on to the next section.

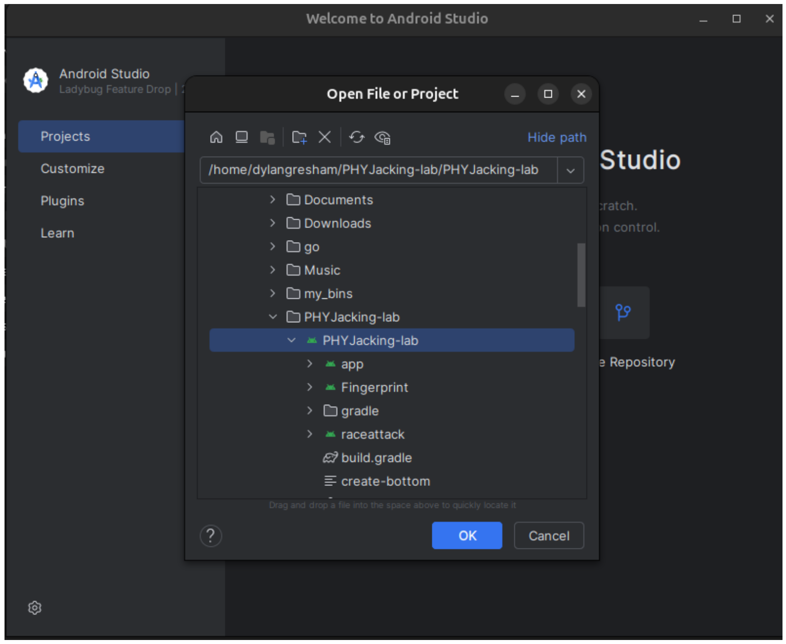

Figure 2: Android Studio open project file picker. Note that the selected directory is the direct parent of all of the source material, not the grandparent directory.

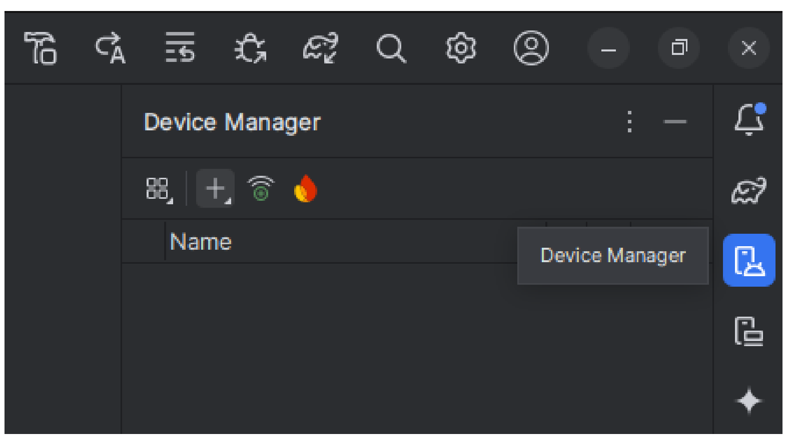

Figure 3: Device Manager menu icon located on the right side of the IDE.

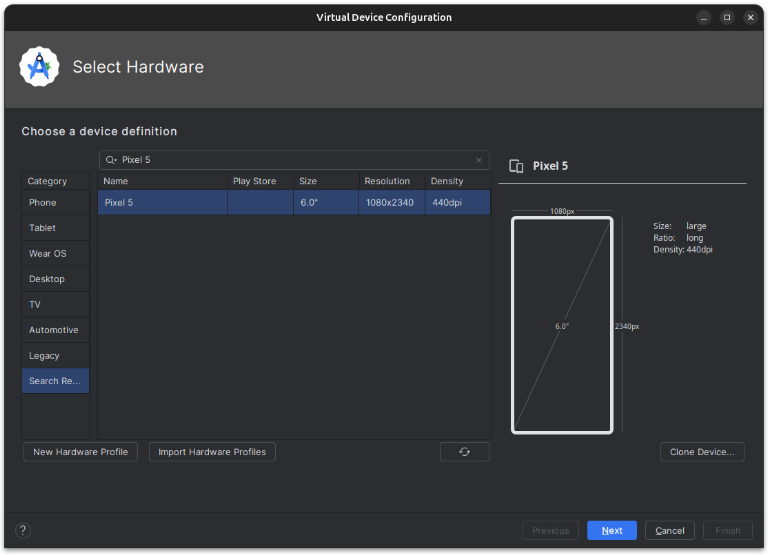

Figure 4: Pixel 5 device selection.

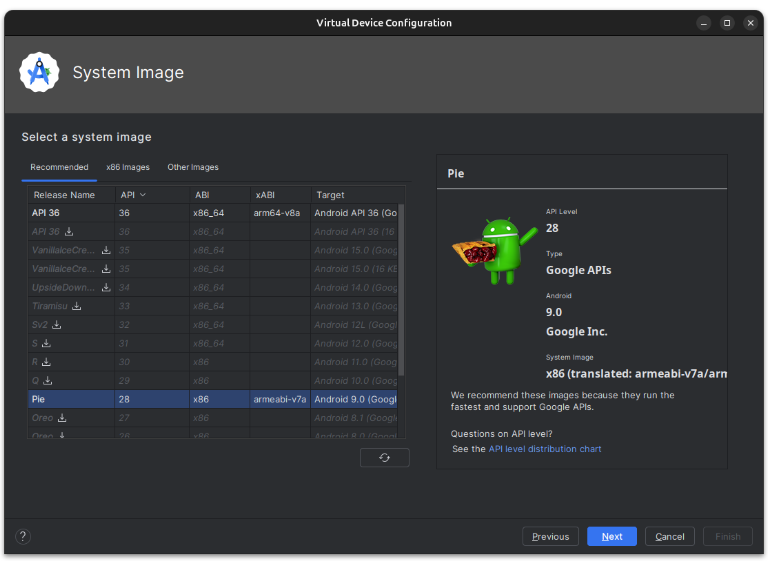

Figure 5: System Image for the Pixel 5. Ensure that Pie with API version 28 is selected.

Pre-made App Installation

1. Start the emulation device from the Device Manager (Figure 6). The first startup may take a few minutes.

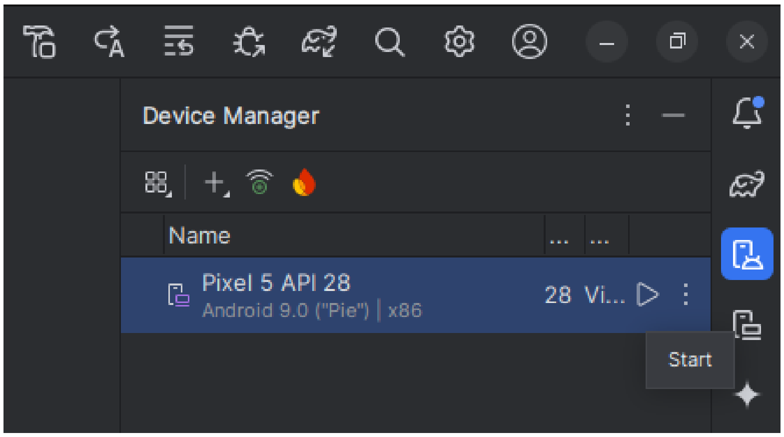

Figure 6: Starting the Pixel 5 emulator.

2. In Android Studio, along the top bar there are run configurations. Select the "raceattack" run configuration and run it (Figure 7). Once the app opens on the Pixel 5 the process can be terminated by either closing the app on the emulator or pressing the red stop button by the run configurations.

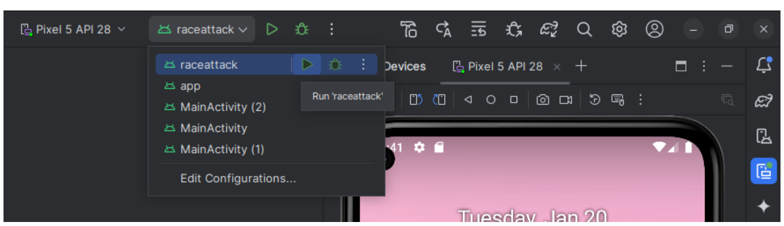

Figure 7: Raceattack run configuration.

3. Repeat the process for the previous step with the "app" run configuration (Figure 8).
4. Once both apps are installed move on to the next section.

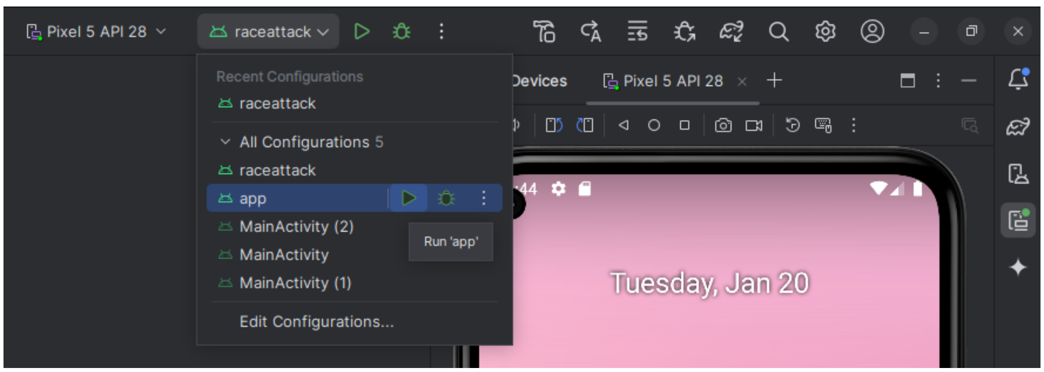

Figure 8: Raceattack run configuration.

Fingerprint Setup

1. In the running emulation device, use your mouse to click and drag up from the home screen to open the app launcher. Open the Settings app, scroll down to the Security & location settings, then open the Fingerprint settings. Refer to Figures 9, 10, and 11 for visual aides.

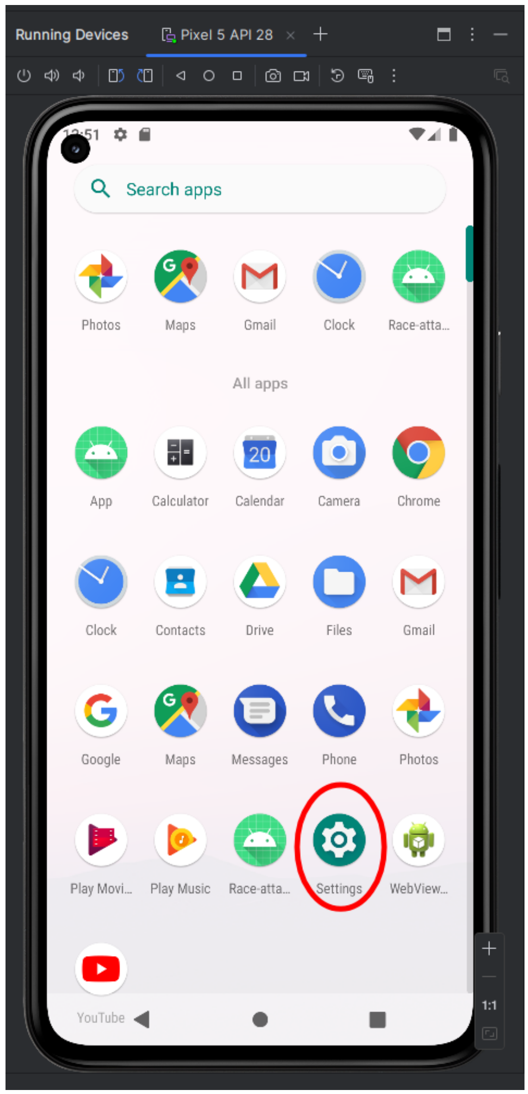

Figure 9: Opening the Settings app.

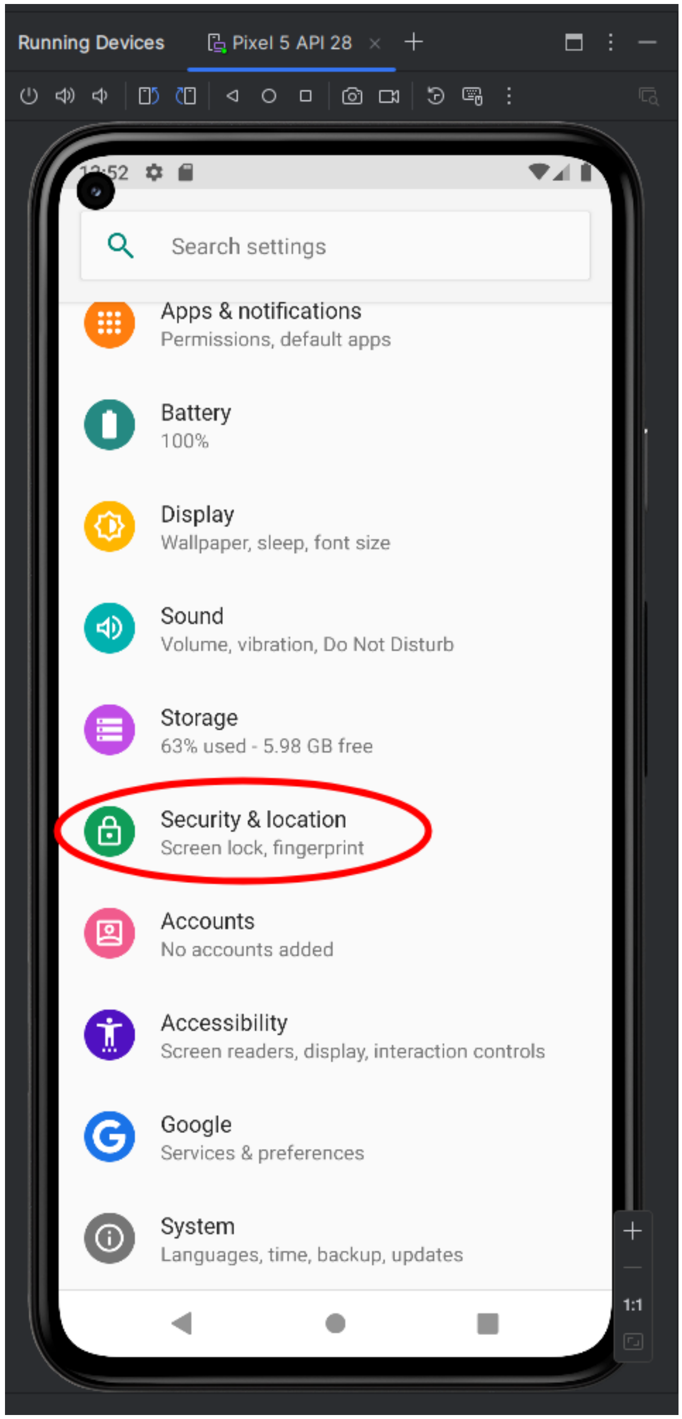

Figure 10: Opening the Security & locations settings.

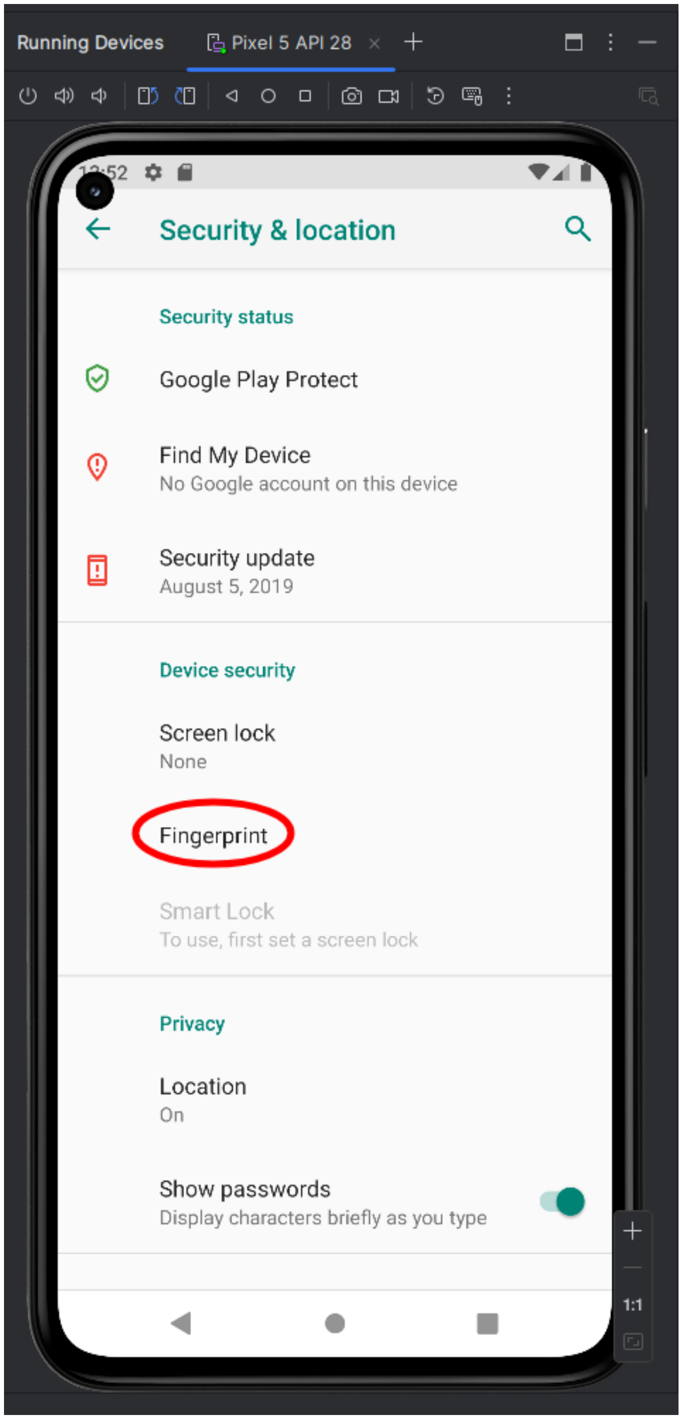

Figure 11: Opening the Fingerprint settings.

2. Choose Next, Fingerprint + Password, Yes, enter "workshop" or any other easy to remember password and select Next, re-type your password and select Confirm, select Show all notification content, Done.

3. To actually use the sensor, select the three dots above the emulation device (Figure 12) to open the Extended Controls menu. Select “Fingerprint” in the Extended Controls Menu (Figure 13) and then click the “Touch Sensor” button until the fingerprint has been added.

4. Congratulations, you’ve successfully set up Android Studio for this workshop and are now ready for the Mobile Security workshop segment of the BEACON Labs Workshop. Thank you for coming prepared! If you haven’t already, please follow the instructions for the Web and Network Security material setups. We look forward to seeing you in Boise for the workshop!

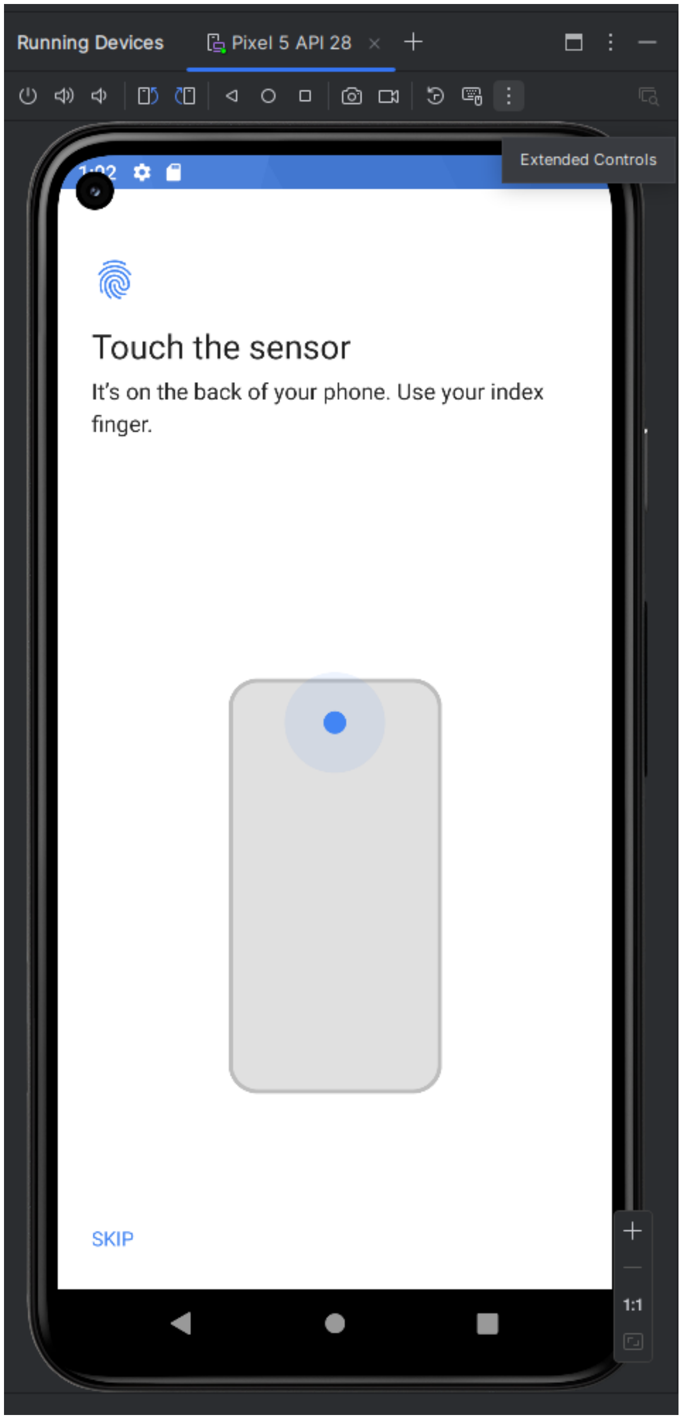

Figure 12: Opening the Extended Controls Menu.

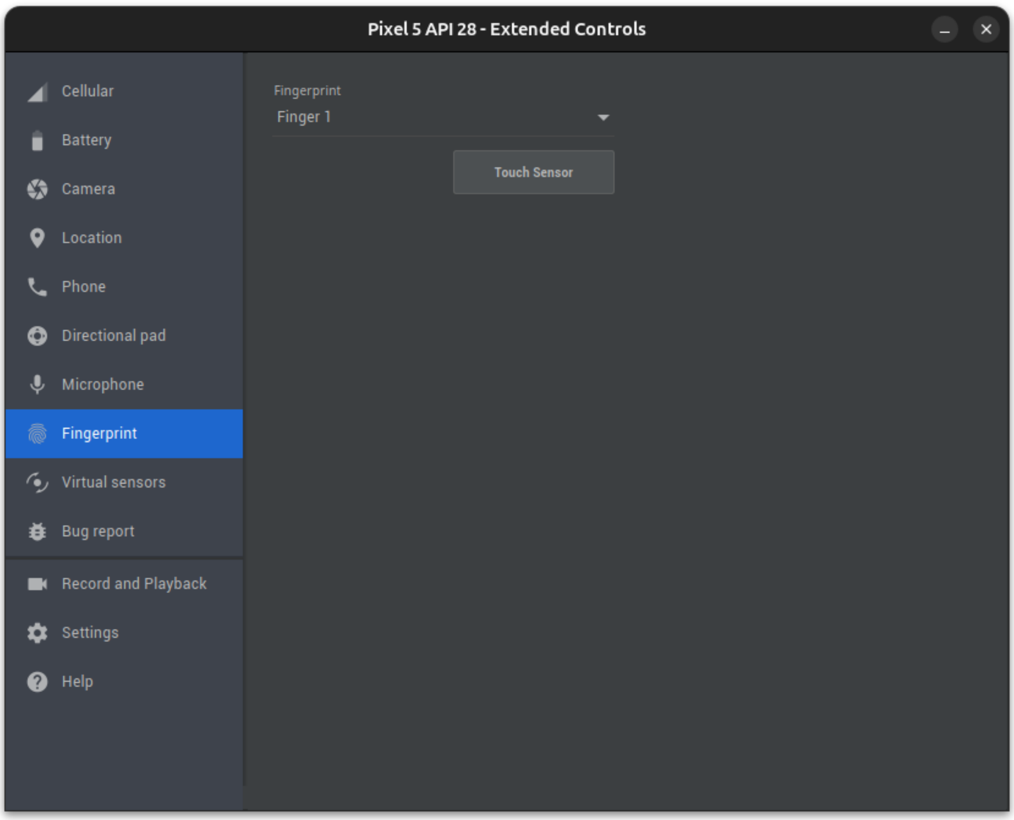

Figure 13: Opening the Fingerprint control menu.